

# ORDI: Orbit-Aware Redundant Distributed Inference for LEO Earth Observation Constellations

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**Abstract**—Low-Earth-orbit Earth-observation constellations collect more imagery than intermittent ground contacts can promptly downlink, motivating distributed onboard inference for time-critical alerts. Yet changing contact geometry, thermal and battery throttling, and correlated orbital-plane failures make static task placement unreliable. We present ORDI, an orbit-aware rolling-horizon scheduler that jointly considers predicted contacts and each satellite’s compute, energy, temperature, queue, and availability state. ORDI ranks image tiles by urgency-weighted utility, selects feasible primary helper-aggregator pairs, and adds fault-disjoint backups only when their marginal reliability gain exceeds energy, communication, and replication costs. Detected failures, missed contacts, and stragglers trigger local replanning. We compare ORDI with eight baselines across fault type and intensity, deadline, constellation scale, correlated outage, and backup-cap studies, using a uniform conservative  $25^\circ$  ground-link elevation mask. Under nominal stochastic loss, ORDI achieves the highest utility while using 40% fewer replicas and 40% less energy than full replication. Under one-plane outages it remains within 3.5 percentage points of full replication’s realized miss ratio with 44% less energy. Its greedy solution attains slightly higher utility than an ILP reference with  $5.8\times$  less inter-satellite traffic. Tests on real-world NASA FIRMS wildfire requests further preserve the synthetic study’s utility ordering, supporting selective redundancy under operational demand conditions.

**Keywords**—LEO constellations, Earth observation, distributed inference, fault-tolerant scheduling

## I. Introduction

Earth observation constellations now produce imagery at rates that overwhelm their ground segments. A single satellite can capture tens of gigabytes per orbit, yet sees a ground station for only a few minutes per pass, and smaller operators often lease a handful of stations concentrated at high latitudes. The result is a widening gap between what is sensed and what can be delivered before it loses value: a wildfire alert is useful within minutes, not hours [1], [2].

Onboard inference attacks this gap at the source. Instead of downlinking raw images, satellites run compact neural networks (e.g., MobileNetV2, YOLOv5-nano) and downlink only labels, bounding boxes, or masks, a reduction of three to four orders of magnitude in volume [3]. Recent in-orbit experiments on the Tiansuan constellation demonstrate that COTS accelerators can run such models in orbit, but also document the operational reality: computing capacity is throttled by thermal limits and battery state, and the energy available per orbit is scarce and variable [4].

A natural next step is to distribute inference: a source satellite partitions an image into tiles and offloads tiles over inter-satellite links (ISLs) to less-loaded neighbors, meeting deadlines that onboard compute alone cannot. Prior systems explore single-satellite query bifurcation [2] and multi-satellite placement [5], but largely assume that scheduled helpers and links remain available. In orbit this assumption routinely fails, and it fails in structured ways: an ISL outage severs all routes through a plane, a plane-wide bus anomaly takes down every satellite sharing it, and thermal throttling turns a scheduled helper into a straggler mid-task. Classic terrestrial answers, replicate everything [6] or apply MDS-coded redundancy [7], [8], translate poorly: every replica costs scarce ISL bandwidth and battery energy, and coding assumes independent failures that orbital geometry violates.

This paper asks: how much redundancy does distributed orbital inference actually need, and where should it be placed? Our answer is ORDI (Orbit-aware Redundant Distributed Inference), a rolling-horizon scheduler built on three ideas.

(1) Orbit-aware feasibility. ORDI plans on a time-expanded contact graph derived from orbital propagation, jointly with per-satellite state: compute rate, battery energy, temperature, queue, and availability. A candidate assignment of a tile to helper  $i$  with aggregator  $a$  is feasible only if its end-to-end latency, transfer to the helper, compute, relay to the aggregator, and downlink, fits the remaining deadline budget under current link and node state.

(2) Selective, fault-disjoint redundancy. Each tile receives one primary replica chosen by marginal utility, and at most one backup, placed only when the marginal gain in delivery probability, discounted by a freshness decay and priced against energy, bandwidth, and an explicit replication penalty  $\lambda_R$ , is positive. Backups must use a different helper and a different aggregator than the primary, and under correlated-failure threat models, a different orbital plane, so that a single structural failure cannot eliminate both replicas.

(3) Reactive replanning. Detected helper failures, missed contacts, and stragglers trigger re-scheduling of affected tiles within the epoch, using the same feasibility machinery.

We implement ORDI and eight baselines in a discrete-epoch simulator with a realistic LEO-EO configuration: a

Walker constellation at 550 km, field-of-view-constrained task arrivals, per-workload log-normal deadlines reflecting EO service tiers [9], and satellite power, thermal, and compute parameters calibrated from public in-orbit measurement logs of a Tiansuan COTS payload [4]. Experiments over up to 60 random seeds show that ORDI (i) outperforms non-distributed baselines on deadline miss ratio and delivered utility, (ii) retains 76% more delivered utility than non-redundant placement at 50% fault intensity, (iii) stays within 3.5 percentage points of full replication’s realized miss ratio under a one-plane outage with 44% less energy, and (iv) slightly exceeds the ILP reference’s utility while using  $5.8\times$  less ISL traffic.

## II. Background and Related Work

### A. Onboard Inference and Orbital Edge Computing

EO satellites are bandwidth-starved: ground contact windows last only 5–10 minutes per pass, and many operators schedule only high-latitude stations, so a satellite over the southern hemisphere may wait most of an orbit to downlink. Onboard inference attacks this at the source, reducing a megabyte-scale tile to a kilobyte-scale result and shifting the bottleneck from downlink to onboard compute, which is in turn bounded by a small spacecraft’s power and thermal envelope. That this is practical is now established by flight programs across agencies and industry: ESA’s  $\Phi$ -Sat-1 flew the first on-board deep-neural-network demonstrator [10], re-trainable payloads have been uplinked and run in orbit on a 1 W VPU [11], the NASA/HPE Spaceborne Computer-2 ran AI workloads on the ISS with software radiation hardening [12], and the Tiansuan constellation reported systematic in-orbit COTS measurements [4] showing accelerators throttled well below nameplate throughput by thermal and battery limits. Earlier work established the paradigm (orbital edge computing [1], onboard model specialization [3]), and recent surveys catalogue it [13]. This body of work is almost entirely single-satellite inference; ORDI addresses the next question of distributing one task across satellites when a single satellite’s compute or downlink cannot meet the deadline.

### B. Multi-Satellite Offloading and Placement

As LEO mega-constellations mature, several systems schedule computation across satellites. Serval bifurcates near-real-time EO queries on one satellite’s resources [2]; SECO optimizes latency or energy for multi-satellite placement [5]; and LEOEdge co-designs satellite-ground AI-inference placement with per-satellite adaptive models [14]. These approaches optimize placement under nominal availability but assume scheduled resources survive; we quantify the resulting reliability loss and show that structured redundancy recovers utility efficiently.

### C. Priced Redundancy under Correlated Failures

Tiling makes inference embarrassingly parallel: a  $4\times 4$  grid yields 16 independent units that ISLs can spread across a neighborhood. But every replica’s success requires the source to emit the tile, each ISL hop to hold, the helper to survive compute, and the aggregator’s downlink to materialize—and these events are correlated, since satellites in one plane share geometry, eclipse cycles, and deployment batch, so a plane-wide outage disables many helpers at once. Tail-tolerant duplication [6] and coded computation [7], [8] address this in datacenters, but assume cheap bandwidth and independent failures. In orbit, full replication costs roughly double the energy and ISL traffic that a 20–30 Wh-class spacecraft cannot afford, and correlation breaks the coding assumption. The design space between zero redundancy and full replication, priced by orbital state and placed against the actual failure correlation, is the territory ORDI explores.

## III. System Model

We consider satellites  $\mathcal{S}$  and ground stations  $\mathcal{G}$ . At epoch  $t$ , the contact graph is

$$\mathcal{C}(t) = (\mathcal{S} \cup \mathcal{G}, E(t)), \quad E(t) = E^{\text{ISL}}(t) \cup E^{\text{DL}}(t),$$

where  $E^{\text{ISL}}(t) \subseteq \mathcal{S} \times \mathcal{S}$  contains satellite-to-satellite contacts and  $E^{\text{DL}}(t) \subseteq \mathcal{S} \times \mathcal{G}$  contains satellite-to-ground contacts.

For a  $d$ -bit payload available at  $a$  in epoch  $t$ ,  $\ell_{ab}(t, d)$  is the minimum delay to  $b$ , including transmission, multi-hop forwarding, and contact waits.  $\ell_{ab}(t, d) = \infty$  if  $b$  is unreachable from  $a$  within the horizon.

Satellite  $i$  has state  $\sigma_i(t) = (C_i, B_i, \Theta_i, Q_i, A_i)$  with domain

$$C_i \in \mathbb{R}_{>0}, \quad B_i \in [0, B_i^{\max}], \quad \Theta_i \in \mathbb{R}, \\ Q_i \in \mathbb{R}_{\geq 0}, \quad A_i \in \{0, 1\}.$$

Here  $C_i$  is available compute throughput (FLOP/s),  $B_i$  is battery energy (J),  $\Theta_i$  is payload temperature ( $^{\circ}\text{C}$ ),  $Q_i$  is committed but unfinished computation (FLOPs), and  $A_i$  is a binary availability flag.

$$\sigma_i(t+1) = F_i(\sigma_i(t), u_i(t), \omega_i(t); \xi_i), \quad (1)$$

where  $u_i(t)$  is the scheduled compute and communication load,  $\omega_i(t)$  collects exogenous conditions such as illumination and faults, and  $\xi_i$  denotes fixed platform parameters. The transition is the component-wise map

$$F_i = (F_i^C, F_i^B, F_i^{\Theta}, F_i^Q, F_i^A),$$

not a learned function:  $F_i^B$ ,  $F_i^{\Theta}$ , and  $F_i^Q$  first account for the epoch’s energy, heat, and completed work;  $F_i^C$  derives the next available rate from the resulting operating state; and  $F_i^A$  applies the battery, temperature, and failure constraints. This abstraction allows the scheduler to consume measured, predicted, or simulated state without depending on a particular hardware model. Section V-A

gives the concrete component equations and parameters used in our evaluation.

Task  $k$ , released at  $r_k$  on source  $s_k$ , is partitioned into tiles  $v$ . Each tile has input and output sizes ( $d_{kv}^{\text{in}}, d_{kv}^{\text{out}}$ ), compute demand  $c_{kv}$  (FLOPs), utility  $u_{kv}$ , and remaining deadline  $\tau_k(t)$ . A replica assigns a tile to helper  $i$  and aggregator  $a$ . Given node, path, and downlink success probabilities, its end-to-end success probability is

$$p_{kvia} = \pi_{s_k} \pi_i \cdot \pi_{\text{path}}(s_k \rightarrow i) \cdot \pi_{\text{path}}(i \rightarrow a) \cdot \pi_{\text{down},a}. \quad (2)$$

Because source survival is shared by every replica, the success probability of replica set  $R_{kv}$  is

$$z_{kv} = \pi_{s_k} \left[ 1 - \prod_{r \in R_{kv}} (1 - p_r / \pi_{s_k}) \right], \quad (3)$$

which does not credit replicas for a shared source failure. We otherwise approximate failures as independent and explicitly test correlated plane outages in Section V-J.

Let  $L_{kv}$  be the earliest completion latency among a tile's replicas. The scheduler selects replica sets to maximize

$$\sum_{k,v} u_{kv} z_{kv} e^{-\alpha L_{kv}} - \lambda_E E - \lambda_C C - \lambda_R R, \quad (4)$$

subject to  $L_{kvia} \leq \tau_k(t)$ , energy reserve  $B_i \geq B_{\min}$ , availability, and link-capacity constraints. Here  $E$ ,  $C$ , and  $R$  denote energy, congestion-weighted ISL traffic, and extra replicas;  $e^{-\alpha L}$  discounts stale results. The resulting routing, placement, and redundancy problem is an integer program; Section IV gives the online heuristic and Section V-K uses the exact program only as a small-instance reference.

#### IV. ORDI Design

Our code is available at <https://github.com/namwoam/ordi>. ORDI is a rolling-horizon greedy scheduler (Algorithm 1). At each epoch it reads the current compute, battery, thermal, queue, and availability state, builds earliest-arrival routing caches from the time-expanded contact graph, and orders pending tiles by urgency-weighted utility  $u_{kv}/\tau_k$ . This ordering gives scarce compute and contact capacity to valuable tiles with little remaining slack.

For tile  $(k, v)$ , an available helper-aggregator pair  $(i, a)$  is a candidate with latency

$$L_{kvia} = \ell_{s_k i}(d_{kv}^{\text{in}}) + \frac{c_{kv}}{C_i(t)} + \ell_{ia}(d_{kv}^{\text{out}}) + \ell_a^{\text{down}}(d_{kv}^{\text{out}}), \quad (5)$$

and is feasible iff  $L_{kvia} \leq \tau_k(t)$ ,  $A_i = A_a = 1$ , and its committed energy leaves the helper above  $B_{\min}$ . The set also includes helper-as-aggregator and source self-processing cases.

We compute  $\ell_{ab}(t, d)$  with Dijkstra over states  $(e, n, b_{\text{rem}})$ : epoch, current node, and undelivered bits. The search starts at  $(t, a, d)$ . Traversing an active contact of rate  $r$  and residual capacity  $q$  sends  $x = \min(b_{\text{rem}}, q)$  bits at cost  $x/r$ ; any residual advances to the next epoch. A

storage edge instead advances one epoch at the same node and charges the contact-wait time. The first zero-residual state popped at destination  $b$  gives  $\ell_{ab}(t, d)$ ; exhaustion gives  $\infty$ . Single-source sweeps fill the source-to-all, all-pairs, and all-to-ground caches once per epoch rather than running one search per candidate.

Each feasible candidate is scored by its primary-placement gain

$$\Delta_{kvia} = u_{kv} p_{kvia} e^{-\alpha L_{kvia}} - \lambda_E \Delta E_{kvia} - \lambda_C \Delta C_{kvia}, \quad (6)$$

where  $\Delta E$  includes receive, compute, and transmit energy and  $\Delta C$  prices ISL bits by inverse link capacity. The highest-scoring candidate becomes the primary. The remaining candidates are scanned in score order; a candidate with a different helper and aggregator (and, under a correlated-failure model, a different orbital plane) is admitted as the backup only if

$$u_{kv} \Delta z_{kv} e^{-\alpha L} - \lambda_E \Delta E - \lambda_R > 0, \quad (7)$$

where  $\Delta z_{kv}$  is the reliability increase from Eq. (3). Thus reliable primaries remain unreplicated, while uncertain paths receive one fault-disjoint backup when its freshness-discounted benefit covers its cost. Committing a replica updates the epoch's energy and link budgets before the next tile is considered.

In Algorithm 1,  $H_R$  is the set of selected helpers and  $\phi(h)$  returns helper  $h$ 's orbital plane. PlaneDisjoint is enabled only for the correlated plane-outage model in Section V-J.  $\Delta(r \mid R)$  is the marginal gain in Eq. (7). The default  $b_{\max}=1$  implements selective single-backup redundancy; Section V-I ablates larger caps. Routing costs  $O(n \cdot \text{SSSP})$  per epoch and enumeration costs  $O(mn^2)$  cached lookups, whereas the ILP is limited to small instances.

#### V. Evaluation

##### A. Setup

Network. Unless stated otherwise, we simulate a 6-plane  $\times$  6-satellite Walker constellation at 550 km (96 min period) for three orbits, discretized into 288 epochs of 60 s. The altitude lies within PlanetScope's 450–580 km Earth-observation range [15]. Optical ISLs operate at 200 Mb/s for slant range  $d \leq 4,000$  km, the specified limit for Starlink mini lasers [16]; at 550 km this is 26% below the 5,408 km geometric line-of-sight limit and leaves 255 km limb clearance. Ground links use a conservative 25° elevation mask, matching an authorized minimum for LEO user and gateway links at comparable altitudes [17]. Two northern stations (Fairbanks and Greenwich) deliberately constrain downlink opportunities.

Tasks and satellite state. Field-of-view-gated Poisson arrivals average six tasks/orbit over 100 ground targets. Four EO profiles use MobileNetV2, YOLOv5-nano, lightweight U-Net, and Siamese CNN for wildfire, ship, cloud, and change detection. Their deadline medians are 300, 600, 1200, and 480 s with log-normal  $\sigma=0.6$ , reflecting

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**Algorithm 1** ORDI rolling-horizon scheduler

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Require: epoch  $t$ , pending tiles  $\mathcal{P}$ , states  $\sigma(t)$ , contacts  $E(t)$ , backup cap  $b_{\max}$ , PlaneDisjoint

- 1: Build earliest-arrival cache  $\Lambda(t)$  from  $E(t)$
- 2: Sort  $\mathcal{P}$  by decreasing  $u_{kv}/\tau_k(t)$
- 3: for  $(k, v) \in \mathcal{P}$  do
- 4:    $\mathcal{F}_{kv} \leftarrow$  feasible replicas satisfying Eq. (5), availability, and energy constraints
- 5:   if  $\mathcal{F}_{kv} = \emptyset$  then
- 6:     continue
- 7:   end if
- 8:    $r_1 \leftarrow \arg \max_{r \in \mathcal{F}_{kv}} \Delta_r$ ;  $R \leftarrow \{r_1\}$
- 9:    $\mathcal{D} \leftarrow \mathcal{F}_{kv} \setminus R$ ; sort  $\mathcal{D}$  by decreasing  $\Delta_r$
- 10:   while  $\mathcal{D} \neq \emptyset$  and  $|R| - 1 < b_{\max}$  do
- 11:     Remove candidates sharing a helper or aggregator with  $R$
- 12:     if PlaneDisjoint then
- 13:        $\mathcal{D} \leftarrow \{r \in \mathcal{D} : \phi(h_r) \notin \phi(H_R)\}$
- 14:     end if
- 15:      $r_b \leftarrow$  first  $r \in \mathcal{D}$  with  $\Delta(r | R) > 0$
- 16:     if  $r_b$  does not exist then
- 17:       break
- 18:     end if
- 19:      $R \leftarrow R \cup \{r_b\}$ ;  $\mathcal{D} \leftarrow \mathcal{D} \setminus \{r_b\}$
- 20:   end while
- 21:   Commit  $R$ ; update energy and link usage
- 22: end for
- 23: if failure, missed contact, or  $1.5\times$  straggler then
- 24:   Mark failed helpers unavailable
- 25:   Reschedule only affected tiles within epoch  $t$
- 26: end if

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EO service tiers [9]. Compute, power, battery, and solar parameters are parsed from public Atlas 200DK measurements on BUPT-1 [4]. From its four-thread inference trace, we compute

$$C_i^{\max} = \frac{N_{\text{img}} F_{\text{img}}}{T_{\text{run}}} = 9.67 \text{ GFLOP/s},$$

where  $N_{\text{img}}$  is the completed image count,  $F_{\text{img}}=0.9$  GFLOP is the profiled work per image, and  $T_{\text{run}}$  is measured execution time. The simulator then produces the available rate as

$$C_i(t) = \eta_i(t) \rho_i(t) C_i^{\max}, \quad \rho_i(t) = \begin{cases} 1, & \vartheta_i < 0.9, \\ 0.60, & 0.9 \leq \vartheta_i < 1, \\ 0.25, & \vartheta_i \geq 1, \end{cases} \quad (8)$$

where  $\vartheta_i = \Theta_i/\Theta_i^{\max}$  and  $\eta_i = 1$  normally or 0.1 during an injected straggler. The thermal factors are conservative stress parameters, not values inferred from the Atlas trace; the source platform uses fixed power levels and showed less than 1% satellite-ground performance variation [4].

After each synthetic scheduling epoch, the simulator aggregates the compute, source-to-helper input, helper-to-aggregator relay, and aggregator-to-ground traffic of newly

committed replicas and calls the transition in Eq. (1) for every satellite, including idle ones. For each epoch of length  $\Delta=60$ s, let  $a_i(t)$  be newly committed work and  $w_i(t) = \min\{Q_i(t) + a_i(t), C_i(t)\Delta\}$  the work executed. Thus  $t_i^c = w_i/C_i$ . ISL transmit, downlink, and receive volumes  $(x_i^{\text{isl}}, x_i^{\text{dl}}, x_i^{\text{rx}})$  produce

$$E_i^x = P_i^{\text{tx}} \min \left\{ \Delta, \frac{x_i^{\text{isl}}}{200 \text{ Mb/s}} + \frac{x_i^{\text{dl}}}{100 \text{ Mb/s}} \right\},$$

$$E_i^r = 0.4 P_i^{\text{tx}} \min \left\{ \Delta, \frac{x_i^{\text{rx}}}{200 \text{ Mb/s}} \right\}.$$

Battery energy is then updated by

$$B_i(t+1) = \left[ B_i(t) + \Delta(P_i^{\text{sol}} - P_i^{\text{idle}}) - t_i^c P_i^{\text{comp}} - E_i^x - E_i^r \right]^{B_i^{\max}} \quad (9)$$

where  $[y]_0^b = \min\{b, \max\{0, y\}\}$ . Because  $P_i^{\text{sol}}$  is the measured orbit-average harvested power, it is applied in every epoch rather than through a separate binary eclipse model. Average compute power is  $\bar{P}_i^c = P_i^{\text{comp}} t_i^c / \Delta$ . The first-order RC update is

$$\begin{aligned} \Theta_i^{\text{ss}}(t) &= \Theta_i^{\text{amb}} + R_i^{\Theta} \bar{P}_i^c(t), \\ \Theta_i(t+1) &= \Theta_i^{\text{ss}}(t) + [\Theta_i(t) - \Theta_i^{\text{ss}}(t)]e^{-\Delta/\tau_i}, \\ Q_i(t+1) &= Q_i(t) + a_i(t) - w_i(t), \end{aligned} \quad (10)$$

where  $R_i^{\Theta}=16.62^\circ\text{C/W}$  is derived from the measured 17–57°C rise divided by 2.41 W incremental compute power, and  $\tau_i=300$ s. The updated temperature determines  $\rho_i$  in Eq. (8); the next available rate is therefore computed after the physical-state update. Finally,

$$A_i(t+1) = \mathbf{1}[\neg f_i(t) \wedge B_i(t+1) \geq 0.15 B_i^{\max} \wedge \Theta_i(t+1) < 80^\circ\text{C}],$$

where  $f_i(t)$  denotes an injected node failure. Initial battery charge is sampled uniformly from 70–95% and initial temperature from 18–35°C. Matched algorithms receive identical initial states and exogenous fault traces; their subsequent state can differ because their assignments impose different compute and communication loads. Table V lists the measured parameters, simulator thresholds, and workload sizes.

Reliability and scheduler. Per-epoch success probabilities are  $\pi_{\text{isl}}=0.97$ ,  $\pi_{\text{down}}=0.92$  (0.70 adverse), and  $\pi_{\text{node}}=0.98$ . We set  $\alpha=0.002 \text{ s}^{-1}$ ,  $\lambda_E=10^{-5}$ ,  $\lambda_C=10^{-12}$ ,  $\lambda_R=0.05$ , and one backup by default. The core comparison further stresses the setup with a sparser  $6 \times 4$  Walker and eight tasks/orbit; it aggregates 60 seeds. Other experiments use matched seeds across algorithms, and error bars show one standard deviation. Table VII gives each experiment’s deviations.

Baselines. B1 direct downlink (bent pipe, no inference); B2 onboard-only; B3 onboard with  $0.15\times$  output compression; B4 Serval-like priority bifurcation, one satellite per task [2]; B5 SECO-like multi-satellite placement without

TABLE I

One-epoch agreement with in-orbit measurements. Battery errors are SOC percentage points (pp).

| State transition       | 60-s intervals | MAE     | RMSE    |
|------------------------|----------------|---------|---------|
| Temperature $\Theta_i$ | 282            | 2.43°C  | 2.89°C  |
| Battery $B_i$          | 168,647        | 0.60 pp | 1.70 pp |

redundancy [5]; B6 full replication to  $r_{\max}$  helpers [6]; B7 random-helper replication; B8 CoCoI-like MDS-coded redundancy adapted to contact windows [7], [8]. All baselines share ORDI’s contact graphs, state model, and routing caches.

Metrics. Deadline miss ratio (primary), delivered utility per (4)’s first term, partial task coverage, recovery latency, ISL traffic, energy, and mean replicas per tile. Energy includes onboard compute, ISL communication, and spacecraft-to-ground transmission; the latter is charged as transmit power times airtime at the modeled 100 Mbps downlink rate, consistent with demonstrated high-rate CubeSat links surveyed by NASA [18]. Thus B1 pays to transmit raw tile inputs, while inference methods transmit compact outputs.

#### B. Agreement with In-Orbit State Data

We check whether the concrete transition model in Eqs. (9)–(10) is consistent with the BUPT-1 data used in our experiments [4]. We first derive its fixed parameters directly from the flight artifact: the inference trace gives  $C_i^{\max}=9.67$  GFLOP/s and idle/active payload powers of 7.78/10.19 W, while the discharge curves give  $B_i^{\max}=118.87$  Wh. These are calibration targets, rather than prediction results.

We then evaluate one 60-s transition at a time, holding the initial measured state fixed. For temperature, the input is the measured active-sample fraction in each non-overlapping interval of the 1-s inference trace. For battery, we apply the idle, orbit-average-harvest update to consecutive samples of the four-month 60-s telemetry trace; SOC is estimated using a conservative 14.4–16.8 V empty/full envelope. Table I reports the resulting agreement. The thermal update is within 5°C for 98.2% of intervals, and the battery update is within one percentage point for 89.6%. However, 89.6% of measured SOC transitions are unchanged at the voltage proxy’s resolution; the battery result therefore bounds short-horizon update error rather than demonstrating detailed charge dynamics.

This check is deliberately short-horizon and in-sample: the long trace lacks synchronized workload, communication, and illumination labels, and its voltage-derived SOC is quantized. We therefore do not claim open-loop trajectory accuracy. Nor do these nominal-temperature data validate the 60%/25% stress throttles, which remain conservative sensitivity parameters. The real-data case study in Section V-L avoids those extrapolations by re-playing measured battery and temperature states directly.

TABLE II

Core comparison results, mean over 60 seeds.

| Algorithm       | Miss ratio | Utility | Replicas | Energy (J) |
|-----------------|------------|---------|----------|------------|
| B1 direct       | 0.951      | 8.0     | 0        | 0.41       |
| B2 onboard      | 0.853      | 17.2    | 0.15     | 45.6       |
| B3 compress     | 0.853      | 17.6    | 0.15     | 4.7        |
| B4 Serval-like  | 0.853      | 17.2    | 0.15     | 45.6       |
| B5 SECO-like    | 0.705      | 22.8    | 0.30     | 94.8       |
| B6 full repl.   | 0.705      | 24.4    | 0.54     | 171.9      |
| B7 random repl. | 0.705      | 23.6    | 0.55     | 175.4      |
| B8 CoCoI-like   | 0.705      | 24.5    | 0.55     | 174.9      |
| ORDI            | 0.702      | 26.0    | 0.32     | 103.0      |

#### C. Core Comparison

Table II and Fig. 1 summarize. Direct downlink misses 95.1% of deadlines, confirming that bent-pipe delivery is not viable with this constrained ground segment. Onboard-only methods (B2–B4) miss 85.3%, limited by source compute and source-local downlink geometry. Distributed schedulers reduce misses to about 70.5%. Within that group, ORDI achieves the highest utility ( $26.0 \pm 14.7$ ) at 0.32 replicas/tile and 103 J. Full replication uses 0.54 replicas and 172 J—about 67% more energy and  $2.5\times$  the ISL traffic—yet delivers less utility (24.4). The sub-unit replica averages include infeasible tiles for which no placement can meet the deadline.

#### D. Robustness Across Fault Types

Injecting each of eight fault classes against ORDI (Fig. 2), ground-contact misses, adverse downlink, battery shortage, thermal throttling, and stragglers leave the planned miss ratio at the no-fault value of 53.3%. ISL disruption raises it to 59.1%; plane outage and helper failure are most damaging at 61.7%. Stragglers do not change feasibility but increase energy from 119 to 218 J as replanned duplicates complete alongside slow originals.

#### E. Graceful Degradation Under Fault Intensity

Sweeping random-fault intensity from 0 to 50% (Fig. 3), B5’s miss ratio rises from 51.2% to 81.2%, while ORDI reaches 77.5%. The miss-ratio gap is modest under the restrictive contact mask, but ORDI retains substantially more value: at 50% faults it delivers 28.5 utility versus B5’s 16.1. Its replica count falls from 0.56 to 0.31 because heavy faulting removes feasible backups. Full replication holds misses near 53% by consuming roughly three times ORDI’s energy at 50% intensity.

#### F. Scalability

From 12 to 60 satellites (Fig. 4), misses fall from 89.8% to 4.0% as contact and helper density improve. ORDI and B5 see the same planned feasibility, so their miss ratios coincide; ORDI delivers 7–16% more utility across the sweep through latency-aware placement and selective backups.

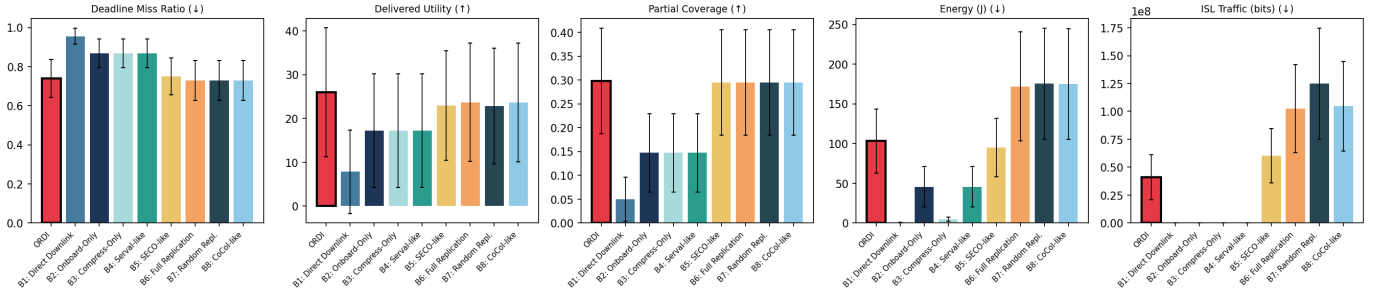


Fig. 1. Core comparison over 60 seeds (no injected faults beyond baseline stochastic link/node loss). ORDI attains the best delivered utility and ties the best miss ratio at near-minimal redundancy.

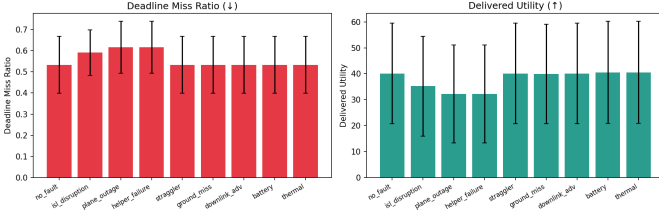


Fig. 2. ORDI robustness profile across eight fault types.

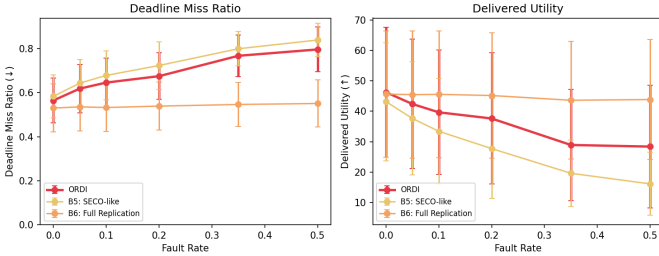


Fig. 3. Deadline miss ratio vs. fault intensity. ORDI degrades gracefully where non-redundant scheduling collapses.

## G. Deadline Sensitivity

Sweeping the global deadline scale from 150 to 900s (Fig. 5), ORDI beats the single-satellite schedulers at every point. It misses 94.2% versus 95.5% at 150s and 55.4% versus 77.1% at 900s; its utility advantage grows from 17% to 60%. Mean replicas rise from 0.09 to 0.48 as longer deadlines make distributed placements and backups feasible.

## H. Pricing Redundancy with $\lambda_R$

Sweeping  $\lambda_R$  from 0 to 2 (Fig. 6) moves mean replicas from 0.48 to 0.30 while the planned miss ratio remains 69.9%. Delivered utility falls from 28.4 to 26.8 and energy from 144 to 91J, showing that  $\lambda_R$  provides a direct confidence–energy trade-off. We use  $\lambda_R=0.05$  throughout, near the transition before the curve saturates at 0.1.

## I. Backup-Cap Ablation

To isolate whether the one-backup cap leaves useful reliability on the table, we set  $\lambda_R=0$ , permit up to four

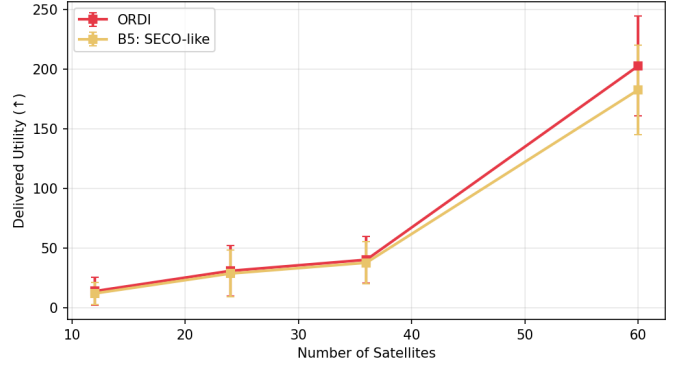


Fig. 4. Scalability with constellation size.

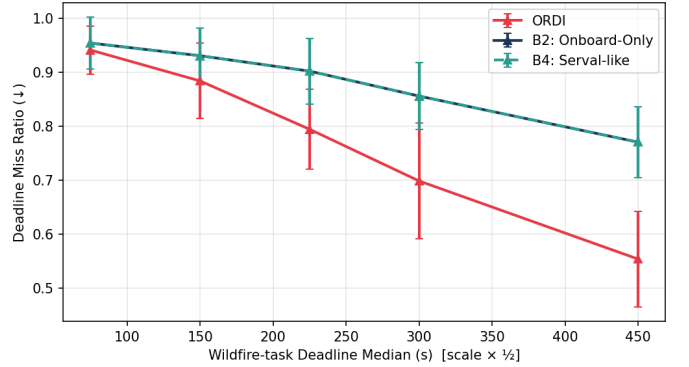


Fig. 5. Deadline tightness sweep.

total replicas, and sweep  $b_{\max}$  from 0 to 3 (Fig. 7). The first backup cuts realized misses from 73.9% to 72.3% and raises utility from 26.8 to 28.4. A second reaches 72.2% misses and 28.44 utility; a third adds no measurable reliability and only 0.003 utility. Energy grows from 91J with no backup to 144, 169, and 179 J. Thus one backup captures 93% of the maximum observed miss reduction, supporting  $b_{\max}=1$  as the default.

## J. Correlated Failures

Independence (Eq. 3) is the standard analytical convenience that orbital geometry breaks. We disable one

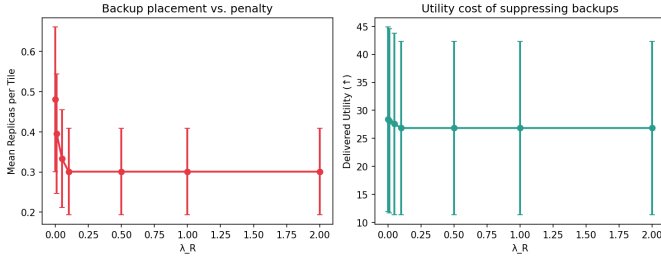


Fig. 6. Replication penalty sweep:  $\lambda_R$  is an effective redundancy dial.

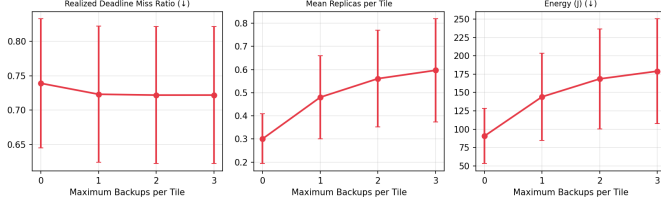


Fig. 7. Backup-cap ablation over 16 seeds with  $\lambda_R=0$ .

or two entire orbital planes mid-horizon (Fig. 8). With plane-disjoint backups, ORDI realizes 59.8% misses for one plane and 61.3% for two, versus 56.3% and 57.7% for full replication. That 3.5–3.6-point gap comes with 44–45% fewer replicas and 44–45% less energy, quantifying the price of the final reliability increment.

#### K. Greedy vs. ILP Optimality

On small instances solvable exactly, greedy ORDI and the ILP reference produce identical planned miss ratios (88.9%). Greedy delivers slightly higher utility (4.49 vs. 4.40) while using  $5.8\times$  less ISL traffic and 32% less energy. The ILP optimizes modeled expected reliability, but its additional redundancy does not improve realized execution in these contact-constrained instances.

#### L. Real-World Wildfire Case Study

Scenario. We replace the synthetic input drivers with three measured datasets: CelesTrak TLEs for 24 Planet Flock/SkySat spacecraft [19], 387 in-view NASA FIRMS active-fire detections [20], and BUPT-1 battery and temperature telemetry [4]. Each detection supplies a real time, location, and fire-radiative-power-derived urgency; deadlines bracket operational wildfire-alert latency [21]. TLEs determine the contact graph, while telemetry replay replaces the RC state model. Eight seeds vary only quantities absent from these datasets: the covering source satellite, tile-level compute jitter, telemetry phase, and reliability outcomes.

Outcome. Table III preserves ORDI’s utility advantage and gives it the lowest planned miss ratio among the reported methods. Full replication lowers the realized miss ratio from 47.2% to 44.9%, but uses  $1.9\times$  the replicas,  $2.0\times$  the energy, and  $13.8\times$  the ISL traffic (2.79 vs. 0.20 Gb). Direct downlink has a 48.4% planned miss ratio.

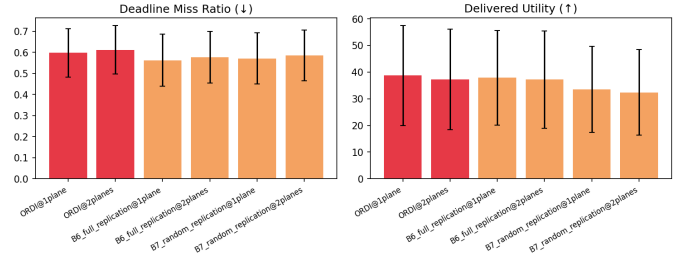


Fig. 8. Correlated plane-wide outages: plane-disjoint backups approach full-replication reliability at lower redundancy.

TABLE III  
Real-world wildfire case study, mean over eight seeds.

| Algorithm     | Miss  | Utility | Energy (J) | Repl. |
|---------------|-------|---------|------------|-------|
| B1 direct     | 0.484 | 1242.0  | 31         | 0     |
| B5 SECO-like  | 0.466 | 614.7   | 254        | 0.53  |
| B6 full repl. | 0.426 | 781.3   | 545        | 1.15  |
| ORDI          | 0.413 | 1311.0  | 279        | 0.60  |

Thus selective redundancy remains effective when orbital geometry, urgent request locations, and platform state all come from operational data rather than synthetic generators.

#### VI. Discussion and Limitations

Fidelity. Our evaluation is simulation calibrated with in-orbit measurements, not a flight experiment. The compute profiles for the four workloads are benchmark-derived; we plan hardware-in-the-loop measurement on Atlas 200DK / Ascend-class accelerators, matching our calibration payload, to replace them and to validate the thermal-throttling model against device behavior. Reliability parameters. Link success probabilities are literature-typical constants; weather-dependent downlink models would sharpen the adverse cases. Coordination. ORDI assumes the source satellite has epoch-fresh constellation state; quantifying staleness tolerance, and a decentralized variant, is future work. Coding. We compare against an MDS-style baseline but do not explore coded partial redundancy (fractional parity tiles), which may dominate replication for large tile counts.

Deployment. ORDI admits two deployment models. Contact windows are deterministic given ephemerides, so a ground operator can precompute epoch graphs and uplink them with tasking; only the per-epoch assignment loop, sub-second at 60 satellites and dominated by dense array lookups after the routing sweeps, must run onboard. Alternatively, the full scheduler can run on the source satellite, since the Tiansuan logs show the assignment workload is negligible next to a single tile inference. Replanning is local by construction: it touches only tiles whose replicas failed, so a recovery decision does not require re-coordinating the constellation. Generality. Nothing in the design is specific to the Walker pattern or the two-station ground segment;

the contact graph abstracts both. Denser ground segments shrink  $\ell^{\text{down}}$  and shift the bottleneck toward compute, which strengthens the case for distribution while leaving the redundancy calculus unchanged.

## VII. Conclusion

ORDI treats redundancy as a priced, orbit-aware resource rather than a blanket policy. One fault-disjoint backup captures 93% of the maximum observed miss reduction in the backup-cap ablation. Under a one-plane outage, selective placement stays within 3.5 percentage points of full replication’s realized miss ratio with 44% less energy, while the greedy scheduler uses  $5.8\times$  less ISL traffic than the ILP reference. Calibration with public in-orbit COTS measurements grounds these conclusions in flight-measured power, compute, and battery behavior.

We see three immediate extensions. First, hardware-in-the-loop calibration: replacing benchmark-derived workload profiles with per-power-mode latency and energy measurements on Jetson-class flight-representative accelerators, validating the thermal-throttling model against observed frequency scaling. Second, coded partial redundancy: fractional parity tiles between one replica and two may dominate both on the energy-reliability frontier for large tile grids. Third, decentralization: bounding how stale a satellite’s view of constellation state can be before selective redundancy loses its advantage, and gossiping only the state that the marginal-gain test actually consumes. Code, declarative fault specifications, and all experiment configurations are available for reproduction.

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## Appendix A Full Simulation Parameters

Table V lists every parameter of the simulator, grouped by subsystem. Satellite hardware parameters are not assumed: they are parsed at runtime from the public Tiansuan BUPT-1 Atlas 200DK in-orbit logs [4] (inference throughput from the 4-thread image-classification log, power from telemetry, battery capacity from published charge curves).



TABLE IV

Complete core-comparison results, mean over 60 seeds. Cvg. = partial coverage; Rec. = recovery latency; Util% = helper utilization; Obj. = modeled objective; Repl. = mean replicas per tile.

| Algorithm       | Miss  | Utility | Cvg.  | Rec. (s) | ISL (Mb) | Downlink (Mb) | Energy (J) | Util% | Obj. | Repl. |
|-----------------|-------|---------|-------|----------|----------|---------------|------------|-------|------|-------|
| B1 direct       | 0.951 | 8.0     | 0.049 | 305.6    | 0        | 8.27          | 0.41       | 0     | 8.0  | 0     |
| B2 onboard      | 0.853 | 17.2    | 0.147 | 682.6    | 0        | 11.91         | 45.6       | 0.001 | 17.2 | 0.15  |
| B3 compress     | 0.853 | 17.6    | 0.147 | 682.6    | 0        | 3.58          | 4.7        | 0.001 | 17.6 | 0.15  |
| B4 Serval-like  | 0.853 | 17.2    | 0.147 | 683.0    | 0        | 11.91         | 45.6       | 0.001 | 17.2 | 0.15  |
| B5 SECO-like    | 0.705 | 22.8    | 0.295 | 721.6    | 60.4     | 25.10         | 94.8       | 0.001 | 22.8 | 0.30  |
| B6 full repl.   | 0.705 | 24.4    | 0.295 | 833.9    | 102.4    | 25.10         | 171.9      | 0.003 | 20.1 | 0.54  |
| B7 random repl. | 0.705 | 23.6    | 0.295 | 878.5    | 124.8    | 25.10         | 175.4      | 0.003 | 19.2 | 0.55  |
| B8 CoCoI-like   | 0.705 | 24.5    | 0.295 | 836.5    | 104.6    | 25.10         | 174.9      | 0.003 | 20.0 | 0.55  |
| ORDI            | 0.702 | 26.0    | 0.298 | 700.3    | 41.0     | 25.17         | 103.0      | 0.002 | 25.6 | 0.32  |

TABLE V  
Complete simulation parameters.

| Group                                                 | Parameter                    | Value                             |
|-------------------------------------------------------|------------------------------|-----------------------------------|
| Constellation                                         | Pattern                      | Walker, 6 planes $\times$ 6 sats  |
|                                                       | Altitude / period            | 550 km / 5760 s                   |
|                                                       | Horizon                      | 17,280 s (288 epochs of 60 B6–B8) |
|                                                       | Ground stations              | Fairbanks, Greenwich              |
|                                                       | GS min. elevation            | 25° (all benchmarks)              |
|                                                       | ISL maximum range            | 4,000 km                          |
|                                                       | ISL rate                     | 200 Mb/s                          |
| Tasks                                                 | Arrival process              | Poisson, 6 tasks/orbit            |
|                                                       | FOV gating                   | 600 km radius, 100 targets        |
|                                                       | Deadlines                    | log-normal, $\sigma=0.6$          |
|                                                       | Tile size                    | 128 $\times$ 128 $\times$ 3 uint8 |
| Scheduler                                             | $\lambda_E$ (energy)         | 10 <sup>-5</sup> utility/J        |
|                                                       | $\lambda_C$ (comm)           | 10 <sup>-12</sup> utility/wt-bit  |
|                                                       | $\lambda_R$ (replication)    | 0.05 utility/replica              |
|                                                       | $\alpha$ (freshness)         | 0.002 s <sup>-1</sup>             |
|                                                       | Replica cap $r_{\max}$       | 2 (primary + 1 backup)            |
| Reliability                                           | $\pi_{\text{isl}}$           | 0.97                              |
|                                                       | $\pi_{\text{down}}$          | 0.92 (0.70 adverse)               |
|                                                       | $\pi_{\text{node}}$          | 0.98                              |
|                                                       | Straggler trigger            | 1.5 $\times$ expected latency     |
| Satellite<br>(measured,<br>Atlas 200DK<br>BUPT-1 [4]) | Compute rate                 | 9.67 GFLOP/s                      |
|                                                       | Idle power                   | 7.78 W                            |
|                                                       | Compute power (incr.)        | 2.41 W                            |
|                                                       | Comms power (incr.)          | 2.49 W                            |
|                                                       | Battery capacity             | 118.87 Wh                         |
|                                                       | Battery floor                | 15%                               |
|                                                       | Solar harvest (avg.)         | 14.43 W                           |
|                                                       | Thermal limit / max obs.     | 80°C / 57°C                       |
|                                                       | Effective thermal resistance | 16.62 °C/W                        |
|                                                       | Thermal RC constant          | 300 s                             |

TABLE VI

Per-tile workload profiles (four EO inference types). D = median deadline at reference scale.

| Type     | Model       | In    | Out   | GFLOP | $u$ | D (s) |
|----------|-------------|-------|-------|-------|-----|-------|
| Wildfire | MobileNetV2 | 48 kB | 1 kB  | 0.3   | 1.0 | 300   |
| Ship     | YOLOv5-n    | 48 kB | 5 kB  | 0.9   | 0.8 | 600   |
| Cloud    | U-Net (lt.) | 48 kB | 50 kB | 1.2   | 0.5 | 1200  |
| Change   | Siamese CNN | 96 kB | 10 kB | 1.8   | 0.9 | 480   |

## Appendix B

## Complete Core-Comparison Results

Table IV reports all ten metrics of the core comparison, mean over 60 seeds (Table II shows the headline columns). B6–B8 lose 4–5 objective points to the replication and congestion penalties of (4), which ORDI largely avoids.

## Appendix C

## Fault Injection Details

Faults are declarative specifications resolved against the built constellation at runtime. The eight classes and their mechanics: ISL disruption removes a specific ISL edge from  $E(t)$  for the fault duration; orbital-plane outage sets  $A_i=0$  for every satellite in one plane; helper failure sets  $A_i=0$  for one satellite, triggering replanning of its assigned tiles; straggler scales the helper's  $C_i$  by 0.1 $\times$  during execution; ground-contact miss removes a downlink window; adverse downlink drops an aggregator's downlink success probability to the adverse value (0.70); battery shortage drains  $B_i$  below the 15% floor, forcing  $A_i=0$  until recharge; thermal throttling raises  $\Theta_i$  past the limit, reducing  $C_i$  to 25%.

The fault-intensity sweep draws, per epoch with probability equal to the fault rate, one fault uniformly from {helper failure, straggler, battery shortage, thermal throttle, ISL disruption} on a random satellite, with durations of 1–3 epochs. The correlated-failure experiment instead disables whole planes at mid-horizon, with backups constrained to plane-disjoint placement.

## Appendix D

## Per-Experiment Configurations

Table VII gives the exact configuration of every experiment. All sweeps hold the Section V-A defaults except the swept variable; matched seeds are used across algorithms within each sweep point.

TABLE VII  
Experiment configurations.

| Exp.              | Swept variable                                  | Algorithms         | Seeds   |
|-------------------|-------------------------------------------------|--------------------|---------|
| Core (V-C)        | none                                            | all 9              | 60      |
| Fault types       | 8 fault classes                                 | ORDI               | 16      |
| Intensity (V-E)   | rate<br>$\{0, .05, .1, .2, .35, .5\}$           | $\in$ ORDI, B5, B6 | 48      |
| Scalability       | $N \in \{12, 24, 36, 60\}$                      | ORDI, B5           | 16      |
| Deadline          | scale $\in \{150..900\}$ s                      | ORDI, B2, B4       | 16      |
| $\lambda_R$ sweep | $\{0, .01, .05, .1, .5, 1, 2\}$                 | ORDI               | 16      |
| Backup cap (V-I)  | $b_{\max} \in \{0, 1, 2, 3\},$<br>$\lambda_R=0$ | ORDI               | 16      |
| Correlated (V-J)  | 1–2 plane outages                               | ORDI, B6, B7       | 16      |
| ILP gap (V-K)     | 9/12 sats, 3/orbit                              | greedy, ILP        | 12 each |
| Real (V-L)        | Planet TLE +<br>FIRMS + telemetry               | all 9              | 8       |